MODULE 01 · EXAM STRATEGY - ALL SECTIONS

The Google Cloud Data Map

Every service the exam names, the 2026 renames, and how to read what a question is really asking.

SECTIONS All five MODULE 1 of 45 SCENARIOS 4

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Most PDE questions are not about what a service does. They are about which one fits **one sentence in the question**.

One platform, five jobs

01

Ingest

Get data in: event streams, files and database changes.

02

Process

Clean, join and enrich, in batch or as a stream.

03

Store

Pick the store that fits how the data will be read.

04

Serve

Dashboards, sharing, machine learning and AI.

05

Govern and operate

Secure, orchestrate, monitor and recover.

MAP TO THE EXAM

Section 2 covers ingest and process, 3 store, 4 serve, 5 operate. Section 1, design, runs across all five.

Getting data in

SERVICE	CHOOSE IT WHEN
Pub/Sub	Events stream in from many producers and you want serverless, global messaging.
Managed Service for Apache Kafka	You already run Kafka, or need Kafka's API and its connectors.
Storage Transfer Service	Files move from another cloud, a list of URLs, or on-premises file servers.
Transfer Appliance	The data is so large that shipping disks beats the network.
Datastream	Change data capture from Oracle, MySQL, PostgreSQL and more.
Database Migration Service	A whole database moves to Cloud SQL or AlloyDB with minimal downtime.
BigQuery Data Transfer Service	Scheduled loads into BigQuery from SaaS apps and other warehouses.

THE SHORTCUT

Ask what is moving: events, database changes, or files.

Processing it

SERVICE	CHOOSE IT WHEN
Dataflow	Batch or streaming pipelines in Apache Beam. Serverless and autoscaling.
Managed Service for Apache Spark (Dataproc)	Existing Spark or Hadoop code, on clusters or as serverless batches.
Cloud Data Fusion	Visual, low-code pipelines with prebuilt connectors.
Dataform	SQL transformations inside BigQuery, in version control, with tests.
BigQuery SQL	The data is already in BigQuery and SQL can do the job.
Cloud Run functions	Small event-driven steps, such as reacting to a file landing.

THE TIEBREAKER

No existing code usually means Dataflow. Existing Spark or Hadoop code means the Spark service.

Storing it

SERVICE	THE PHRASE THAT GIVES IT AWAY
BigQuery	Analytics with SQL over very large data. The warehouse.
Cloud Storage	Files and objects of any size. The lake and the landing zone.
Bigtable	Huge read and write volumes by key, in single-digit milliseconds.
Spanner	Relational, strongly consistent, and scaling out, even globally.
Cloud SQL	Managed MySQL, PostgreSQL or SQL Server at regional scale.
AlloyDB	PostgreSQL-compatible, with fast analytics on live transactions.
Firestore	Serverless document database for application data.
Memorystore	In-memory cache, answering in under a millisecond.

REMEMBER

Module 19 turns these phrases into a full decision tree. For now, learn the phrase next to each name.

Governing and operating it

SERVICE	WHAT IT IS FOR
IAM and organisation policies	Who can do what, and which settings are allowed at all.
Cloud KMS	Encryption keys you control for data at rest, known as CMEK.
Sensitive Data Protection	Finding, masking and tokenising personal data. Formerly Cloud DLP.
Knowledge Catalog (Dataplex)	Discovery, data quality and lineage across the estate.
Managed Service for Apache Airflow (Composer)	Scheduling and orchestrating multi-step pipelines as DAGs.
Workflows	Lightweight orchestration of API calls and serverless steps.
Cloud Monitoring and Cloud Logging	Metrics, alerts, logs and audit trails.

ON THE EXAM

Sections 1 and 5 live here: least privilege, and knowing before the customer does.

Larkspur Energy, end to end

Datastream
Oracle billing CDC
50 tables

Pub/Sub
meter-readings topic
About 1,700 a second

Spark service
Hadoop history, 1.2 PB
Nightly settlement

Dataflow
Dedupe and validate
Windows, late data

BigQuery warehouse
148.8m readings a day
50 billing tables
UK and Germany
Layers: raw, curated, marts

Bigtable
meter_live table
App reads in ms

Knowledge Catalog
Discovery, quality
Lineage

Dashboards
Data Studio for teams
Looker for the board

BigQuery sharing
Westgrid demand feed
Hearthly clean room

REMEMBER

Do not memorise this now. By the end of the course you will be able to defend every box on it.

Meet Larkspur Energy

- **3.1 million smart meters** in the UK and Germany, each sending a reading every half hour.
- **Signal drops.** A meter buffers its readings for up to 72 hours, then uploads them late.
- **An Oracle billing system** and an eight-year **Hadoop history** to migrate by June 2027.
- **Personal and bank data**, and German customer data that must stay in Germany.
- **Two partners**: a grid operator wanting data within five minutes, and a thermostat company.

> REMEMBER
>
> Larkspur is fictional. Its numbers are fixed for the whole course, so you can check one module against another.

In April 2026 Google renamed much of its data stack. The exam guide **still uses the old names**.

Older name, current name

OLDER NAME	CURRENT NAME	CHANGED
Dataproc	Managed Service for Apache Spark	April 2026
Cloud Composer	Managed Service for Apache Airflow	April 2026
BigLake	Google Cloud Lakehouse	April 2026
Dataplex	Knowledge Catalog	April 2026
Vertex AI	Gemini Enterprise Agent Platform	April 2026
Looker Studio	Data Studio, its original name	April 2026
Cloud DLP	Sensitive Data Protection	Earlier
Analytics Hub	BigQuery sharing	Earlier

ON THE EXAM

Same products, same features. Answer with whichever name the question uses.

What the rename did not change

- **APIs, client libraries and the CLI.** Commands such as gcloud dataproc keep working.
- **IAM role names.** Roles such as roles/dataproc.editor keep their old spelling.
- **The exam answer.** A question saying Dataproc wants the same answer as one saying Managed Spark.
- **The trap:** picking an option because its name sounds newer. A new name is not a new product.

> ON THE EXAM
>
> For Dataproc, BigLake and Dataplex, Google states the API, CLI and IAM names are unchanged.

Retired, or on the way out

PRODUCT	STATUS	USE INSTEAD
Data Catalog	Phased shutdown from 1 June 2026	Knowledge Catalog
Pub/Sub Lite	Turned down 18 March 2026	Pub/Sub, or Managed Kafka
gsutil	Legacy. Leaves the gcloud CLI package after March 2027	gcloud storage
Cloud Functions	Renamed Cloud Run functions, August 2024	Cloud Run functions

ON THE EXAM

If an older question's answer is Data Catalog, the idea still holds. Map it to Knowledge Catalog.

Words that eliminate options

THE QUESTION SAYS	WHAT IT RULES OUT
Least operational overhead	Anything you patch, size or keep running: VMs, long-lived clusters.
Minimal code changes	Rewrites. Keep the engine you already run, such as Spark or Kafka.
Most cost-effective	Capacity that sits idle between spiky or occasional jobs.
Near real time, within seconds	Batch loads and scheduled jobs.
Google-recommended practice	Clever custom code where a managed feature exists.
Without moving the data	Copies and exports. Think external tables, Omni, sharing.
Low-code or no-code	Hand-written Beam or Spark. Think Data Fusion.

ON THE EXAM

Find the qualifier before you read the options. It usually removes two of them.

Two qualifiers that pull opposite ways

WHEN IT SAYS

Least operational overhead

- Serverless first: BigQuery, Dataflow, Pub/Sub
- Serverless Spark rather than clusters
- Managed features over custom code
- Accept a rewrite if it removes servers

WHEN IT SAYS

Minimal changes, fast migration

- Keep the engine you already run
- Spark and Hive jobs on Managed Spark
- Kafka clients on Managed Kafka
- Modernise later, once you are out

THE TIEBREAKER

If a question carries both, the deadline usually wins. Migrate first, modernise second.

How the worked scenarios work

- **Pause** when you hear 'pause and decide', and commit to a letter.
- **Find the deciding sentence** in the situation before you weigh the options.
- **Listen to why the wrong answers are wrong.** That is what transfers to new questions.
- **Watching without answering** feels like learning and teaches almost nothing.

> REMEMBER
>
> Four scenarios close this module. They use only what you have seen so far.

> Westgrid, Larkspur's grid partner, needs total demand per region for each half hour, within five minutes of the half hour closing. Readings land on a Pub/Sub topic at about 1,700 a second, rising sixfold every evening.
>
> The team has no one on call for infrastructure and must choose the option with the least operational overhead.

What should the team build?

A. A Spark Streaming job on a long-running Dataproc cluster that reads the topic

B. A streaming Dataflow pipeline that reads the topic, aggregates by region and half hour, and writes to BigQuery

C. A Cloud Run function per message that updates running totals in Cloud SQL

D. A Kafka Streams application on GKE, fed by Managed Service for Apache Kafka

◆ PAUSE AND DECIDE

What should the team build?

A Spark Streaming on a long-running Dataproc cluster

Works, but a long-running cluster is yours to size, patch and keep alive through the evening surge.

B Streaming Dataflow into BigQuery

Serverless, autoscales through the surge, handles time windows natively, and writes straight to the warehouse.

C A Cloud Run function per message into Cloud SQL

About 150 million calls a day, and Cloud SQL becomes the bottleneck for constant updates.

D Kafka Streams on GKE with Managed Kafka

Adds a second messaging system and a cluster to run. Kafka fits when you already have Kafka.

THE SIGNAL

'Least operational overhead' plus a streaming workload points to Dataflow almost every time.

> Larkspur's platform team keeps runbooks full of commands such as gcloud dataproc clusters create, and IAM policies that grant roles/dataproc.editor to a pipeline service account.
>
> A new engineer reads that Dataproc is now called Managed Service for Apache Spark, and raises an urgent ticket: every script and policy must be rewritten before the next release.

What is the correct response to the ticket?

A Move the scripts to the new CLI command group, and keep the existing IAM roles

B Keep the scripts, but replace the IAM roles with the new Managed Spark roles

C Migrate to the new product, because Dataproc clusters stop running after the rename

D Change nothing. The rename did not change the API, CLI or IAM names

◆ PAUSE AND DECIDE

What is the correct response to the ticket?

A Move the scripts to a new CLI command group

There is no new command group to move to. gcloud dataproc is still the CLI.

B Replace the IAM roles with new Managed Spark roles

The roles kept their names. You would be granting roles that do not exist.

C Migrate, because Dataproc clusters stop running

Nothing was migrated or switched off. It is the same service under a new product name.

D Change nothing

Google states the API, client library, CLI and IAM names are unchanged.

THE SIGNAL

Rename questions test whether you treat a new name as a new product. It is not.

> Larkspur's customer app shows each customer their latest meter reading. It must answer in single-digit milliseconds for any of 3.1 million meters, while readings are written at about 1,700 a second, and around 10,000 a second in the evening peak.
>
> A prototype on Cloud SQL slows down every evening.

Which storage service should hold the live readings?

A Bigtable, with the meter ID leading the row key

B BigQuery, with the table clustered on the meter ID

C A larger Cloud SQL for PostgreSQL instance with an index on the meter ID

D Cloud Storage, with one object per meter overwritten on each reading

◆ PAUSE AND DECIDE

Which storage service should hold the live readings?

A Bigtable, meter ID leading the row key

Built for heavy writes and millisecond reads by key. Leading with the meter ID spreads writes across nodes.

B BigQuery, clustered on meter ID

Superb for scanning history. Not built to fetch one row in a few milliseconds.

C A larger Cloud SQL instance

Buys time, but the evening write surge is exactly where the prototype already struggles.

D One Cloud Storage object per meter

Object reads take tens of milliseconds, and every write replaces a whole object. A lake, not a database.

THE SIGNAL

Single-digit milliseconds, lookups by key, and a heavy write rate: that is the Bigtable sentence.

> Larkspur runs about 300 Spark and Hive jobs on a 40-node Hadoop cluster in its Bristol data centre. The lease ends on 30 June 2027, nine months away.
>
> The team wants to stop managing HDFS and hardware, and must keep code changes to a minimum to hit the date.

Which migration approach fits best?

A. Rewrite the jobs as Dataflow pipelines, then switch off the cluster

B. Translate the jobs to BigQuery SQL and run them with Dataform

C. Copy the data to Cloud Storage and run the existing jobs on Managed Service for Apache Spark

D. Recreate the Hadoop cluster on Compute Engine VMs, with HDFS on persistent disks

◆ PAUSE AND DECIDE

Which migration approach fits best?

A Rewrite as Dataflow pipelines

Perhaps a good long-term home for some jobs. Rewriting 300 in nine months is not a minimal change.

B Translate to BigQuery SQL with Dataform

Same problem. Translating everything is a modernisation project, not a migration.

C Cloud Storage plus Managed Service for Apache Spark

Jobs run largely as they are, Cloud Storage replaces HDFS, and clusters can be created per job.

D Hadoop on Compute Engine VMs

Minimal code changes, but you still manage hardware, just virtual. The question wanted that gone.

THE SIGNAL

'Minimal code changes' plus 'stop managing HDFS' means Cloud Storage plus the Spark service. Modernise later.

Module 01 in five lines

- Five jobs: **ingest, process, store, serve, govern and operate**.
- Ingest by what moves: **events, database changes, or files**.
- **New pipelines** go to Dataflow. **Existing Spark** goes to the Spark service.
- The April 2026 renames changed **names, not products**. Use the question's name.
- **Find the qualifier first.** It usually removes two options.

> COMING UP
>
> Next: module 02, IAM and organisation policies for data.